NZ ARMY
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AI and Army Operational Effectiveness

A first-principles analysis for senior Army leadership

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Question	How can Army use AI to increase operational effectiveness?

The Question

> How can Army use AI to increase operational effectiveness?
> What should Army be able to do better because AI exists?

1. This paper reasons from Army's outputs backwards to the technology. It does not assume Army must adopt AI, and it does not catalogue uses. It asks where AI could materially improve Army's ability to generate and employ land capability, what must remain human, and what that means for Army's people.

2. AI is new within the NZDF, but it is not ungoverned. Defence holds a closed enterprise instance of Copilot approved for restricted and protected material, and DFY 6020 directs the use of approved tools only, with responsibilities for labelling and review. The evidence for military effect nonetheless remains thin, and is drawn largely from larger armies with different problems. Several propositions below are working judgements reasoned from Army's situation rather than findings, and are marked as such. They are offered to be tested, not accepted.

1. The First-Principles Answer

3. Army exists to generate land forces ready for operations, to employ them, and to learn from doing both. Almost everything else supports those three outputs.

4. The working judgement of this paper is that for a small army the constraint limiting all three is skilled human capacity. There are not enough experienced people; much of their time goes on staff work that consumes judgement without exercising it; depth is thin; and adaptation is slow because those who must adapt are already fully committed. If that judgement is wrong, the priorities below change.

5. It follows that AI increases operational effectiveness only where it relieves something genuinely limiting an Army output. For a small army, skilled human capacity is likely the most important such limit now.

> Use AI where it removes a real limit on what Army can do, and where that makes Army more capable.

6. Three distinctions govern everything that follows. **Genuine military advantage** changes what Army can do on operations, or how fast it can generate capability. **Administrative efficiency** frees time, and becomes advantage only when leaders redirect the time. **Technological novelty** is neither, and is the category most AI proposals fall into.

2. The Framework

7. One framework organises this paper and tests any proposed application. It reasons from the output to the tool, and closes on measured effect.

Output → Task → Constraint → AI → Human Role → Assurance → Competence → Effect

8. In use:

   a. **Output.** What must Army deliver, and to what standard?

   b. **Task.** Which task limits that output?

   c. **Constraint.** What limits the task: time, quality, scale, understanding or people?

   d. **AI.** Would AI relieve that constraint, and how? If you cannot say how, stop.

e. **Human role.** Augment, automate or enable, and what stays with the person.

f. **Assurance.** What does the consequence of error require by way of control and checking?

g. **Competence.** Who must be able to do what, and how will they practise it?

h. **Effect.** Measure the effect on the output, not the use of the tool. Scale, adjust or stop.

9. A commander, staff officer or training lead should run this on a single page. If you cannot say what is limiting the task, and how AI would help, stop. The sections that follow apply it.

3. Where AI Creates Military Advantage

10. Five areas survive the test, ordered by the strength of the mechanism and by how much of the opportunity Army itself controls. None has been measured inside Army: these are the areas to trial first, not demonstrated findings.

Planning and staff work

11. The appreciation, course of action development, orders production and the routine staff estimate are language and structure tasks consuming the scarcest people in a headquarters. AI can generate and stress-test options, draft and check products against intent and format, and surface omissions, returning staff time to judgement. The decision, the acceptance of risk and the intent remain the commander's. The risk is a plausible product wrong in a detail no one checks, and a staff that loses the ability to run the process unaided.

Understanding: intelligence, information and the state of the force

12. A small intelligence cell cannot process the open-source, coalition and sensor information available to it. AI can fuse, summarise, translate and flag change across volumes a person cannot read. Much of that opportunity is enabled by allies; Army's advantage lies in competent consumption, not in building the system. Separately, readiness data on people, training, equipment and health sits in disconnected systems, seen late and in part. Army cannot currently see itself whole, and the constraint is data quality rather than technology.

Generating trained people

13. Training design, content maintenance, scenario generation, individual practice, feedback and assessment support are the heart of force generation, and where the training approach and AI meet. AI can make adaptive pathways practical at scale, keep content current as tactics and equipment change, and shift instructor effort from transmission to coaching. The gain is readiness, through faster and more adaptive generation of competence. Section 6 develops this.

Learning and adaptation

14. Army's lessons cycle runs on documents and committees. The working judgement here is that it is slower than the tempo at which threats, tactics and equipment now change; Army should test that against its actual cycle times. If it holds, AI can compress the path from observation and after-action review to updated training and doctrine. Institutional learning speed matters for a small army because it cannot afford to relearn. The uncertainty is whether the organisation will act at the speed the tool allows.

Administrative and organisational load

15. Correspondence, reporting, returns, minutes and routine staff process will yield the largest and earliest gains in hours, and are the safest to automate. Per paragraph 6, this is efficiency, and becomes advantage only where commanders convert the hours into training, planning and readiness.

16. Two areas are deliberately not prioritised. Autonomous targeting and weapons are not Army's to develop, and carry unresolved legal and ethical questions. Predictive logistics and personnel analytics are constrained by data maturity and, for personnel, by bias and trust; they follow the data foundations rather than lead them.

Area	Mechanism	Mode	Consequence of error	Human role
Planning and staff work	Faster planning; broader options; staff time returned to judgement	Augment	Medium to high	Decides, accepts risk, verifies
Intelligence and readiness	Scale; understanding sooner, from more	Augment; enable for readiness	High for intelligence; medium for readiness	Assesses sources, judges, decides
Generating trained people	Adaptive competence at scale; readiness	Augment and enable	Low to medium	Coaches, assesses, assures the standard
Learning and adaptation	Speed of institutional learning	Enable	Low to medium	Validates, decides what changes
Administrative load	Human capacity	Automate	Low	Owns what is signed

4. What Should Remain Human

17. The following remain with people, not because AI cannot contribute but because responsibility cannot be delegated to a system:

    a. The decision to use force, the acceptance of risk, and the statement of intent.

    b. Assessment of a soldier's readiness, character and leadership, and the coaching relationship that develops them.

    c. Legal and ethical judgement, including the law of armed conflict and rules of engagement.

    d. Accountability. A commander who signs a product owns it; the tool that drafted it does not. AI-assisted output is official information, and is labelled and reviewed as DFY 6020 requires.

    e. The foundational soldier and staff skills Army must retain as a capability in their own right, because on operations the tools will be degraded, denied or deceived.

18. Step f of the framework is settled by a single test:

The more it matters if AI gets it wrong, the more human control and checking is required

19. Where error is cheap, reversible and easy to check, greater autonomy is acceptable. Where consequence is high, effects are irreversible, or output cannot readily be verified, greater control and assurance are required. This asks the same question as the decision rule in the training approach: match control to what is at stake.

5. What This Means for Army People

20. AI competence is becoming part of military competence, not a specialist skill beside it. Prompt technique is a small part of it. Competence is the ability to employ AI against real Army tasks with judgement, within the security and accountability those tasks require.

21. Four complementary requirements, not a ladder. Everyone understands it; many employ it; leaders govern it; a few specialise in it.

    a. **All personnel.** What these tools are, how they fail, what may and may not be put into them, and the obligation to verify before use.

    b. **Practitioners: junior leaders, staff officers and instructors.** Frame a task so a tool can help; evaluate and verify output critically; integrate tools into real workflows such as orders, planning and training design; know when not to use them.

    c. **Leaders and commanders.** Set risk appetite by consequence; assure outputs; protect their people's underlying competence; own accountability for AI-enabled products; understand what a system cannot tell them. A governance requirement, not a more advanced technical one.

    d. **Specialists.** Data, security and model assurance, and integration of approved tools into Army systems. Few, and shared across the NZDF.

22. Confidence and competence come from practice against real tasks, not from exposure. Investment in tools without practised people has not produced productivity in the wider economy, and will not produce capability in Army.

6. What This Means for Individual Training

23. The training approach holds. AI is an enabler of it, not a change to it. Tested against the approach:

a. **Operational competence.** For many roles the standard now includes the competent, judged use of approved tools. For foundational and safety-critical skills it is unchanged, and assured without the tool.

b. **Common standard, adaptive pathway.** AI makes adaptive pathways practical at scale, through learning data, individual practice and feedback. On this analysis it may be the most significant contribution AI can make to the training system, and the one Army most fully controls.

c. **Progression.** Tools are introduced where they will be used, not at the foundation. Core skills are built without them; tools enter at guided and independent practice for the tasks that will employ them. Instructors must be competent with a tool before their students are, which makes instructor development the first investment.

d. **Continuous development.** AI supports learning before, during and after the course, and shortens the path from feedback to reinforcement when tactics change. Instructor effort shifts from transmitting knowledge to coaching, verifying and assuring.

e. **What remains human, and preventing dependency.** Judgement under pressure, leadership, field skills, the ethical decision, and the trust on which teams are built. Dependency is prevented by design rather than prohibition: foundational skills are practised and assessed tool-denied, with the discipline Army applies to navigation without GPS.

7. Risks and Constraints

24. The risks that matter most, in order of consequence:

a. **Plausible error.** Fabricated facts, references and figures in products that read as authoritative. Mitigated by verification as a duty, never optional where consequence is high.

b. **Security and classification.** Sensitive information entering tools not approved for it; sovereignty over where models run; allied caveats. DFY 6020 addresses the first by directing approved tools only. The others remain open, and rightly limit early use.

c. **Loss of competence.** Staff and soldiers who cannot perform unaided when the tool is denied. Mitigated through training, not policy.

d. **Adversarial manipulation.** Poisoned data, injected instructions and deception aimed at AI-enabled processes. Poorly understood, and growing with dependence.

e. **Bias and trust,** especially in decisions about people. Requires a human decision that can be explained.

f. **Accountability drift.** Decisions made because the system said so.

25. The constraints appear institutional rather than technical: external resources blocked, slow security approval, data in legacy systems, no single owner accountable for converting AI into military advantage, procurement measured in years against a capability that changes in weeks, no spare capacity to experiment, and a risk appetite that defaults to no. An approved enterprise tool is already in place; if this reading is right, Army's ability to exploit what it has matters more than what becomes available next.

8. Principles

26. Five principles for responsible and effective employment:

a. **Effect first.** AI is adopted where it improves an Army output, and is measured against that output.

b. **Task before tool.** Reason from the output and its constraint to the application, never from the technology to a use for it.

c. **The commander owns it.** AI advises, drafts and analyses; where consequence is high it does not decide. Verify what you use. You own what you sign.

d. **Assurance by consequence.** More human control where getting it wrong matters more.

e. **Competence before dependence.** Never depend on AI for a skill Army must be able to perform without it. Competence comes from practice against real tasks, and is measured like any other.

9. Priority Actions

27. Five actions, in order:

a. **Own it.** Assign accountability for AI in Army and set a risk appetite by consequence, so that low-consequence use on approved tools is enabled by default rather than blocked by default.

b. **Prove effect in three places.** Bounded, measured trials: planning and orders production in a headquarters; training design and feedback in Army Training Group; and readiness data integration. Measure effect on the output. Stop what does not work.

c. **Train it through the approach.** A Defence AI course is running; the step now is to build that competence into individual training against real Army tasks, with verification and tool-denied assurance of core skills. Instructors first.

d. **Improve the data.** Readiness and training data may constrain some of the highest-value applications Army can control directly. Test that before investing in it.

e. **Compress the lessons cycle.** Move from observation to updated training and tactics at a pace the force can act on.

10. Unresolved Questions

28. The following remain open, and the analysis above should be read with them in mind:

a. Whether skilled human capacity is in fact Army's binding constraint, on which the priorities here depend.

b. Where Army's models and data may run, under whose sovereignty and classification, including on deployed and coalition networks.

c. How operational effect, rather than tool use or hours saved, will be measured.

d. New Zealand's policy position on AI in targeting and lethal decision, and its coalition obligations.

e. How loss of underlying competence will be detected before it matters.

f. Who does this work in an army of Army's size, and what it displaces.

g. What the AI competence standard is, by rank and trade, and who sets it. A Defence course is running; a course is not a standard.

What Should Army Be Able to Do Better Because AI Exists?

> Army should be able to plan faster with more options genuinely considered; see its own readiness and its environment sooner and more completely; generate trained people more adaptively, along different routes to the same standard; learn from experience and change its training and tactics at a pace the force can act on; and return the time of its most experienced people to judgement, leadership and training. It should be able to do all of this while remaining able to fight when the tools are degraded, denied or deceived. A small army that cannot do the latter has not gained effectiveness. It has moved its dependence.